5 Whys's

Date of 5 why's: **6/27/2024** Update of 5 Why's:

PROBLEM DESCRIPTION

Problem description: **pin of tin on the soldering pad, risk of contamination**

OCCURENCE

Relevant factors from FTA	1st WHY?	1st Check	2nd WHY?	2nd Check	3rd WHY?
1.method of mannuel soldering	a litlle pin of tin was spilled o, soldering pas	>	At the end of soldering, the operator passes the soldering iron (with the tin remaining on the nozzle of the soldering iron) over the BC	>	the work instruction does not contain the desc removing the soldering iron in the end of opera the operator just finish)

1 Result(s)

NON DETECTION

Relevant factors from FTA	1st WHY?	1st Check	2nd Check	2nd WHY?	3rd WHY?
2.Soldering inspection Method	The tin control operator didn't recognize the smal burr of tin	>	>	The operator didn`t have the clean records to manipulate the point.	The control operator doesn`t specify the manipulate fa no longer having a tin in sector of PCBA --> the factor is

1 Result(s)

Nidec Comment

Lesson Learned Card

LLC Type

Date of LLC: **6/26/2024**

Logistics: ☐ Management: ☐

Product: ☐ Detection: ☑

Occurence: ☑ Process: ☐

1. What was the problem?

What happened?

pin of tin on the soldering pad, risk of contamination

Who detected it?

operator

Where was it detected?

CM4 line

How was it detected?

visually

Why is it a problem?

pin of tin on the soldering pad, risk of contamination

When did it happen?

1/30/2024 12:00:00 AM (your local time)

How many parts ?

1.00

2. What is the cause?

At the end of soldering, the operator passes the soldering iron (with the tin remaining on the nozzle of the soldering iron) to be cleanned and the smal burr stick and drop in the sector of PCBA

3. Description of the applied modification

BEFORE	AFTER

Manuel soldering
defect not recognize by control operator

IC700_Cm4__Old_one_ (1).pdf

automatic soldering with robots
updated the control instruction

IC700_Control_soldering_station_updated (1).pdf

4. What we learned?

ERADICATION

What is the countermeasure that was put in place to avoid recurrence ?

For occurence
*/switch to automatique soldering with robot
For non-detection
*/Updated the Control instuction of station control & inspection soldering points
*/training opeartor of control& inspection of soldering point regarding the new issue

CHECKING

What is the method (who, what, how) for the countermeasures?

Who : AVO
What :
*For occurence :
--Follow up the automatic soldering operation
*/For nin detection
-- checking the apply of standard of control

SIGNALING

What is the countermeasure su

In case of detection of NOK part
soldering pad, risk of contamina
line : alert and block all stock fo
+ Alert and block the parts in tra

5. What has been modified?

a. Proces is modified according to Before/ After sheet:	☐	g. Visual aids are posted:	☑
b. All documents have been updated and has been validated with the operators:	☑	h. Operators and other functions are trained:	☑
c. Procedures and work instructions:	☑	i. Proof of training is avalaible (records,5 ' meeting):	☑
d. Control Plan and PFMEA:	☐	j. Operators can correctly describe what they must do:	☐
e. Maintenance instructions:	☑	k. Other:	☐
f. Records (start-up checklists, reaction sheets, preventive maintenance...):	☐		

An evidance of modification:
IC700_Cm4__Old_one_ (1).pdf
TABLEAU_GENERIQUE_CM3CM4_SONIA_DRIDI (1).xls
Automatic_soldering_with_Robots.pptx
IC700_Control_soldering_station_updated (1).pdf
20240628122945192.pdf

6. Potential applicability

LLC Applicability Product: **Brush mounting plate assy**
LLC Applicability Line / process: **CM4**
LLC Applicability Plant: **AVO**

Sorting Approval

Approval of sorting / rework

This document must be sent to Nidec within 24 hour(s) after claim notification

Date of page: Update:

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IMMEDIATE ACTIONS REQUIRED AT NIDEC

PROBLEM DESCRIPTION

pin of tin on the soldering pad, risk of contamination

DESCRIPTION OF SORTING/REWORK